PHY101: Introduction to Physics-1

Monsoon, 2017

Tutorial-3

1. Two particles, *A* and *B*, move with constant velocities v_1 and v_2. At the initial moment their radii vectors are equal to r_1 and r_2. How must these four vectors be interrelated for the particles to collide?

2. Two swimmers leave point *A* on one bank of the river to reach point *B* lying right across on the other bank. One of them crosses the river along the straight line *AB* while the other swims at right angles to the stream and then walks the distance that he has been carried away by the stream to get to point *B*. What was the velocity u of his walking if both swimmers reached the destination simultaneously? The stream velocity v_0 = 2.0 km/hour and the velocity v' of each swimmer with respect to water equals to 2.5 km per hour.

3. A particle of mass *m* is subject to a force $F(t) = \alpha t^3$ and moving along *X*-direction. The initial position and speed are x_o and v_0, respectively. Find *x*(*t*).

4. A particle moves outward along a spiral. Its trajectory is given by r = Aθ, where A is a constant. θ increases with time according to $\theta = \alpha t^2/2$ where α is another constant. Show that the radial acceleration is zero when $\theta = 1/\sqrt{2}$ rad.